Spacescape

Overview: The game of Spacescape still has the main core concepts of the player climbing upwards through platforms while also avoiding asteroids to reach the mothership. I did implement some changes from the design document.

- **Finite Platforms and Scoring:** There is now a fixed number of platforms that are generated, 50, with the final platform followed by the mothership. This changed from the original idea of an infinite game or target score goal because generating the entire game at once made it easier for me to place the mothership and create a clear ending without extra platforms. I also think that having a dedicated end goal with the mothership is more satisfying. This change also affected scoring because the final score combines the final height reached, asteroids destroyed, and then a time bonus based on the remaining countdown time if the game is completed. This is great because it rewards progress and speed.
- **Platforms:** I was able to successfully implement the platforms, which now get smaller and further from each other as the game progresses. Additionally, I made it so that the moving platforms move faster as the game progresses. There are also springs on some platforms that allow for players to be bounced up and skip numerous platforms.
- **Shooting:** The player now has an upward aiming and firing arc instead of only being able to shoot upward, with the angles being about 20 degrees to 160 degrees. There are now two animations divided into aiming direction: straight upwards, from 67.5 degrees to 112.5 degrees, and diagonal, which covers the rest of the available firing arc. These angles support idle, running, and jumping. Even though the animation does not work for every single angle in that arc, the diagonal animation is used across most of that arc. I think this was a good design implementation instead of limiting the player from only shooting upward. There is also a new powerup implemented, OneShot which allows players to destroy asteroids in one shot while it is active.
- **Presentation:** Presentation now resembles the space theme, and there are now dedicated screens including title, playing, game over, and win screens. There are also instructions, score, timer, and powerup availability as part of the interface. There are also dedicated animations for idle, running, jumping, and shooting animations. Asteroids now rotate at random speeds while falling, making it appear more natural.